11	**(a)**		differentiate to obtain $2x - 4$	**B1**	**3.1a**		
			$+ 1 \times \ln x + x \times \frac{1}{x}$ oe	**M1** **A1**	**2.1** **1.1**	use of Product Rule all correct	allow one error
			derivative = 0 oe seen and terms combined $2x - 3 + \ln x = 0$ www **isw AG**	**A1** **[4]**	**2.4**		

Question			Answer	Marks	AOs		Guidance
11	(b)		any rearrangement to obtain $x = g(x)$ from given derivative = 0	M1*	2.1	allow sign error	eg $x = e^{2x-3}$
			$x = \frac{3 - \ln x}{2}$	A1	1.1	any correct rearrangement	need not see subscripts in iterative formula
			use of their $g(x_n) = \frac{3 - \ln x_n}{2}$ to obtain at least two iterates eg 2, 1.1534, 1.4286…	M1dep*	1.1	must see iterates	
			1.3500 cao	A1	2.2a		**0** for 1.3500 unsupported
				[4]			
			…… *Alternatively,*				trial and improvement does not score
			$x_{n+1} = x_n - \frac{2x_n - 3 + \ln x_n}{\text{their}\left(2 + \frac{1}{x_n}\right)}$	M1*	2.1	Newton-Rapshon iterative formula seen (not for solving $f(x) = 0$)	need not see subscripts in iterative formula
			use of their N-R formula to obtain $x_1, x_{2,\ldots}$	A1	1.1	formula all correct	
			eg 1.5, 1.34795, 1.34996,….1.349962	M1dep*	1.1	must see iterates	
			1.3500 cao	A1	2.2a		
				[4]			